AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Final Examination
Spring Semester 2024/2025

Date: 29 /May/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **100**
- This exam consists of **5** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [20 Marks]

For each question, choose the most appropriate answer. Copy the answer to your answer booklet. Each question carries 2 marks.

1. Which of the following returns an integer value (not a floating-point number)?
   a. `Math.floor(1.4)`
   b. `Math.sqrt(1)`
   c. `Math.round(1.4)`
   d. All answers

2. Which of the following is a Java reserved word?
   a. `statice`
   b. `long`
   c. `float`
   d. All answers

3. In the below code, which line has error?

```
1 for (int i=1; i<=10; i++)
2   System.out.println(i);
3 i=i+4;
```

   a. 1
   b. 2
   c. 3
   d. There is no error

4. If `array` is an initialized 2D array, the element in the second row and second column is:
   a. `array[0][0]`
   b. `array[2]`
   c. `array[2][2]`
   d. `array[1][1]`

5. What is the returned value for the following method if it is called using the statement `tryMethod(1)`?

```
public static void tryMethod(int x) {
   System.out.print(x++)}
```

   a. 1
   b. 2
   c. 0
   d. Nothing is returned

6. What is the value stored in `b[0]` after the executing the following Java statement?
   `boolean [] b = new boolean [3];`
   a. `true`
   b. `false`
   c. 2
   d. It is not initialized

7. Which of the following is primitive type in Java?
   a. `int`
   b. `char`
   c. `boolean`
   d. All answers

8. What will be printed after executing the following code:
   `System.out.println(!(20!= 4*5)?"Correct":"Wrong");`
   a. `Correct`
   b. `Wrong`
   c. `true`
   d. `false`

9. The while loop in Java executes at least once, even if the condition is false.
   a. True
   b. False

10. The if statement in Java can be used without an else block.
    a. True
    b. False

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Question 2 [20 Marks]

Trace the following pieces of code and write their exact output:

1. (4 Marks)

```
for (int i = 1; i < 3; i++) {
   for (int j = 1; j <= 2; j++)
       System.out.println( i + j);
}
```

2. (4 Marks)

```
int x = 1;
do
{
  x++;
  if (x>5)
      continue;
  System.out.println(x);
} while (x <= 10);
```

3. (6 Marks)

```
public class Exam {
    public static void main(String[] args) {
        int a = 10, b = 5;
        calculate(a, b);
        System.out.println(a);
        System.out.println(b);
    }
    public static void calculate(int x, int y) {
        x--; y++;
        int z = x+y*2/x-2;
        System.out.println(z);
    }
}
```

4. (6 Marks)

```
int[] x = {2,1,0,3,6,5};
x[0]=x[2+3];
x[1]+=10;
x[2] = x[5] - x[4];
x[3] %= x[4];
x[4]=x[5]--;
for (int i = 0; i <= 5; i++)
   System.out.println(x[i]);
```

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Question 3 [15 Marks]

Consider the following Java code that starts by declaring and initializing a 2D array `grades` representing the grades of 4 students in 3 quizzes. Complete the code to do the following:

1. Find the highest grade.
2. Calculate the average of the first quiz for the 4 students.

```
int[][]grades = {{12, 15, 12},
                 {19, 20, 18},
                 {18, 19, 17},
                 {14, 19, 14}
                };
int highest;

//Write here the code to find the highest grade
.....
.....

System.out.println("The highest grade is:"+ highest);

int sum;
double average;

//Write here the code to calculate the average of the first quiz
.....
.....

System.out.printf("The average of the first quiz:%.2f",average);
```

For questions 4 and 5 in the next page, copy the below code into your answer booklet and continue the rest of the code according to the given specifications.

```
import java.util.Scanner;
public class FinalExam {
      public static void main(String[] args) {
          Scanner input = new Scanner(System.in);

    }
}
```

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Question 4 [20 Marks]

Write Java code to do the following:

1. Repeatedly prompts the user to enter a word ("Quit" to stop).
2. Prompt the user to enter an integer (call it index).
3. Print the letters of the word entered in step 1, starting from the beginning up to the given index (not including the letter at the index).
4. If the index given is negative or bigger than the length of the word, print error message.
5. When the user enters "Quit", stop the program and show how many words were entered in total (not counting "Quit").

Here is a sample of the program run to help you understand how the program works:

```
Enter a word or 'Quit' to stop: Artificial
Enter the index: 3
Art
Enter a word or 'Quit' to stop: Robots
Enter the index: 10
Error: Index out of range
Enter a word or 'Quit' to stop: Intelligence
Enter the index: 5
Intel
Enter a word or 'Quit' to stop: Quit
The total number of words you have entered is: 3
```

Question 5 [25 Marks]

Write Java code to do the following:

1. Write a method called `countMultiplesOf()` that takes 2 arguments: an array of integers, and an integer number(divisor). The method should return the number of elements in the array that are divisible by the given divisor (i.e., the remainder is zero when the element is divided by divisor).
2. In the main method, create a one-dimensional array of 10 integers.
3. Initialize the array with 10 integers entered by the user.
4. Read an integer representing the divisor.
5. Call the method `countMultiplesOf()` with the 2 required arguments that you have read in step 3 and 4 and print the result.

Here are 2 samples of the program run to understand how the program works:

Program run 1:

```
Enter 10 integers: 20 6 7 4 12 3 15 100 95 11
Enter a divisor: 5
There are 4 numbers in the array that are divisible by 5
```

Program run 2:

```
Enter 10 integers: 4 6 12 14 18 9 7 15 10 21
Enter a divisor: 3
There are 6 numbers in the array that are divisible by 3
```

End of questions

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